RETAIL SALES ANALYTICS PROJECT

PHASE 1: DATA PREPARATION (Excel)

1. Convert the Date column to proper date format, then create columns for Month Name, Quarter, and Weekend Flag (Sat/Sun).
2. Create a formula column verifying that Quantity × Price per Unit = Total Amount. Apply conditional formatting to highlight any mismatches (there shouldn't be any!).
3. Create an Age Group column categorizing customers as: Young (18-25), Millennial (26-35), Gen X (36-45), Boomer (46-55), Senior (56-65).
4. Add a column classifying products as "Budget" ($25-$50), "Mid-Range" ($300), or "Premium" ($500).
5. Build a pivot table showing Total Revenue, Transaction Count, and Average Order Value by Product Category.

PHASE 2: POWER BI DATA MODELING

6. Import the Excel file and create a Date Dimension table with: Date, Month, Quarter, Year, Weekday, and Month Name.
7. Set up relationships between your Sales table and Date dimension. Explain why you chose your relationship direction.
8. Create DAX measures for: Total Revenue, Total Transactions, Average Order Value (AOV), and Unique Customer Count.
9. Create a YTD Revenue measure and a Month-over-Month Growth % measure.
10. Create a measure showing each category's revenue as % of total (must work with filters).
11. Create a measure that ranks Product Categories by revenue (1st, 2nd, 3rd) dynamically.

PHASE 3: DASHBOARD VISUALIZATION

12. **KPI Header:** Design a header with cards showing: Total Revenue, Total Transactions, AOV(Average Order Value), and Best Performing Category.
13. **Trend Line:** Build a line chart showing monthly revenue trends with drill-down capability to daily view.

14. Create a clustered column chart showing Revenue by Category with AOV(Average Order Value) as a line overlay.
15. **Demographics,** Design a section with: (a) Donut chart for Gender split, (b) Bar chart for Revenue by Age Group.
16. Create a scatter plot: Price per Unit (X) vs Quantity (Y), with bubble size = Revenue and color = Category.
17. Add interactive slicers for: Date Range, Product Category, Gender, and Age Group.
18. Build a matrix showing Category vs Month with conditional formatting (green for above-average cells, red for below).
19. Create a measure that generates a chart title like "Revenue for [Category] - [Month]" that updates based on slicer selections.

PHASE 4: INSIGHTS & ANALYSIS

20. **Weekend vs Weekday:** Calculate which generates higher revenue per transaction and higher total volume. Visualize the comparison.
21. **Top Customer Segment,** Identify which Age Group + Gender combination has the highest Average Order Value. Would you target them for a premium product launch?
22. Analyze quarterly performance. Which quarter is strongest? Is there a December holiday effect?
23. Do customers buy more units when prices are lower? Compare $25/$30/$50 (Budget) vs $300/$500 (Premium) quantity patterns.

PHASE 5: FINAL DELIVERABLES

24. **Executive Summary:** Create a one-page summary with: (a) Key finding about best category, (b) Customer demographic insight, (c) One actionable recommendation with data proof.

Checklist

1. Cleaned Excel file with engineered columns
2. Power BI (.pbix) with data model + dashboard
3. One-page Executive Summary (PDF/Word)
4. ✅ 5-minute presentation explaining your key insight